Why Do Astronauts Float Around in Space?

Satoshi Furukawa aboard the ISS - Image via NASA

THIS IS A great question. It comes up quite often. If you ask the people around you, there are two common answers

Astronauts float around in space because there is no gravity in space. Everyone knows that the farther you get from Earth, the less the gravitational force is. Well, astronauts are so far from the Earth that gravity is so small. This is why NASA calls it microgravity.

In space, no one can hear you scream. You know why? Because there is no air in space. No air, no sound. No air, no gravity. Simple.

Yes, both of these are wrong. But why?

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Is the gravitational force too weak in space?

What is the gravitational force? It is an interaction between objects that have mass. The Earth has mass and the astronaut has mass - so they are attracted. We can model this attractive force with the following expression.

$$F_{\text{grav}} = G\frac{M_1 m_2}{r^2}$$

In this model for the magnitude of the gravitational force, M_1 and m_2 are the masses and r is the distance between the centers of these two masses. G is the gravitational constant. It has a value of 6.67 x 10^{-11} N*m^2/kg^2. Oh, but what about the famous (or infamous) g = 9.8 N/kg? (or commonly listed in units of m/s2) That value is just for objects on the surface of the Earth. Check this out. If I have something sitting on the ground, it interacts with the Earth. The mass of the Earth is 5.97 x 10^{24} kg and the center of the Earth is 6.38 x 10^6 m away (the radius of the Earth). Let me put these values into the gravitational model.

$$F_{\text{grav}} = (6.67 \times 10^{-11}\ \text{Nm}^2/\text{kg}^2)\frac{(5.97 \times 10^{24}24\ \text{kg})}{(6.38 \times 10^6\ \text{m})^2}m_2$$

$$F_{\text{grav}} = (9.78\ \text{N/kg})m_2$$

Yes, that is not 9.8 N/kg. I used rounded values in the calculation so that it is off just a bit. But you get the idea. I am getting off track here. Doesn't this expression say that the gravitational force gets weaker as you get farther from the Earth? Yes. But not by has much as you think. A typical height for an orbiting Space Shuttle is about 360 km above the surface of the Earth. Suppose I have a 75 kg astronaut. What would be the weight (gravitational force) on the astronaut both on the surface and in orbit? The only difference will be the distance between the astronaut and the center of the Earth.

$$F_{\text{grav-surface}} = (6.67 \times 10^{-11}\ \text{Nm}^2/\text{kg}^2)\frac{(5.97 \times 10^{24}\ \text{kg})(75\ \text{kg})}{(6.38 \times 10^6\ \text{m})^2}$$

$$F_{\text{grav-surface}} = 734\ \text{N}$$

And in orbit:

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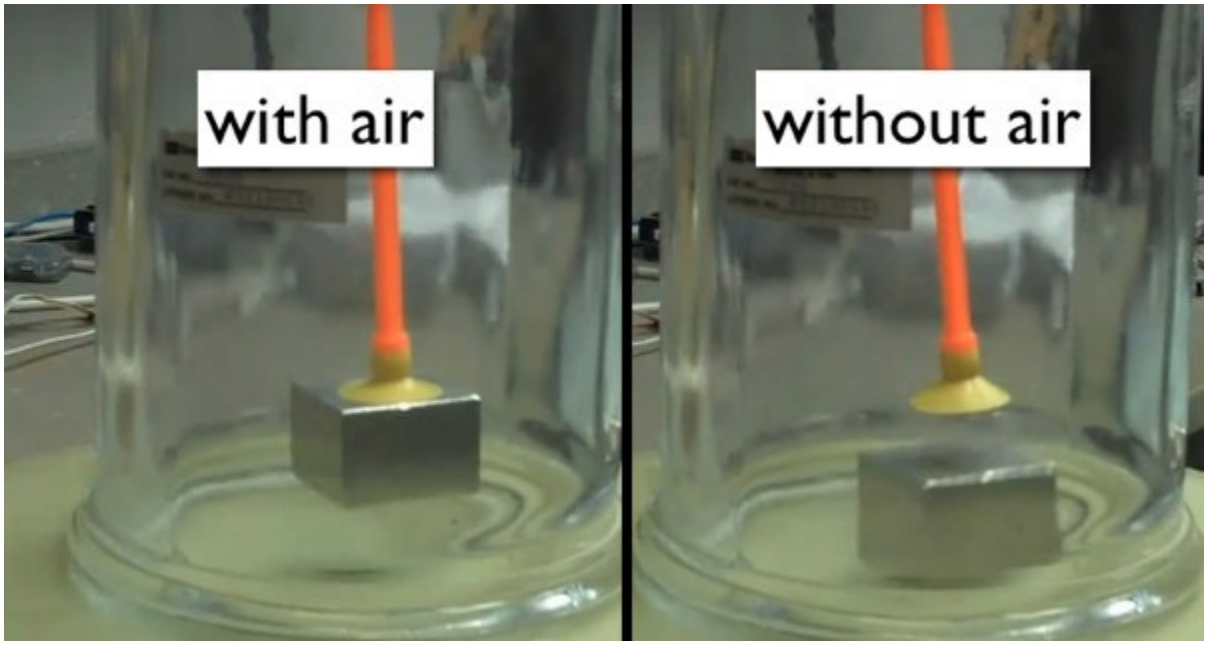
$$F_{\text{grav-orbit}} = (6.67 \times 10^{-11}\ \text{Nm}^2/\text{kg}^2)\frac{(5.97 \times 10^{24}\ \text{kg})(75\ \text{kg})}{(6.38 \times 10^6\ \text{m} + 3.6 \times 10^5\ \text{m})^2}$$

$$F_{\text{grav-orbit}} = 657\ \text{N}$$

Smaller? Yes. Enough to call it “weightless”? No. The gravitational force in orbit is 89% as large as on the surface. So, this isn’t the correct explanation for “weightlessness”.

What about the lack of air?

You can probably find some examples of why this isn’t the cause of “weightlessness”. Here is one that I like. Basically, it is a demonstration of how a suction cup works. I made a video of a mass hanging from a suction dart inside a vacuum bell. (link here) This is a picture of the mass before the air was pumped out.



And what about the Earth itself? Why does it orbit the Sun? It orbits because there is a gravitational force between the two objects. There is an interaction even though there is no air between them.

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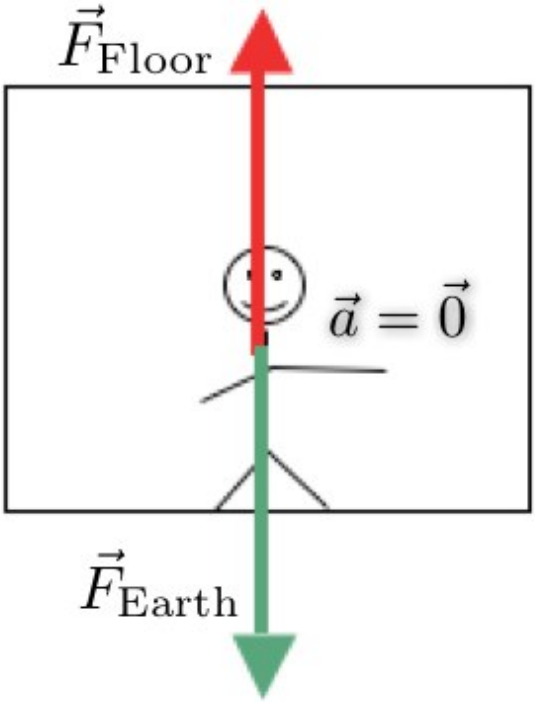
Then why do you float?

Maybe I should talk about how you feel weight. What is your apparent weight? Let me go ahead and say that what you are feeling right now isn't really gravity. Suppose I start with some examples.

Example 1: Go stand in an elevator. Do not push the buttons. Just stand there so that the elevator is at rest. How do you feel? Awkward? Here is a diagram.



Since you are at rest and staying at rest, you are in equilibrium (acceleration is zero). If your acceleration is zero, the net force must also be zero (technically, the zero vector). The two forces on you are the force from the floor pushing up and gravitational interaction with the Earth pulling down. The magnitudes of these two forces have to be equal in order for the net force to be zero.

Example 2: Now push the "up" button. During the short interval that the elevator accelerates upwards, how do you feel? Anxious? Or maybe you feel a tad bit heavier. If your elevator is like the one in this building, you might feel frustrated at how slow the damn thing goes. And what's that funny smell? Here is a diagram for the upward accelerating elevator (and you).

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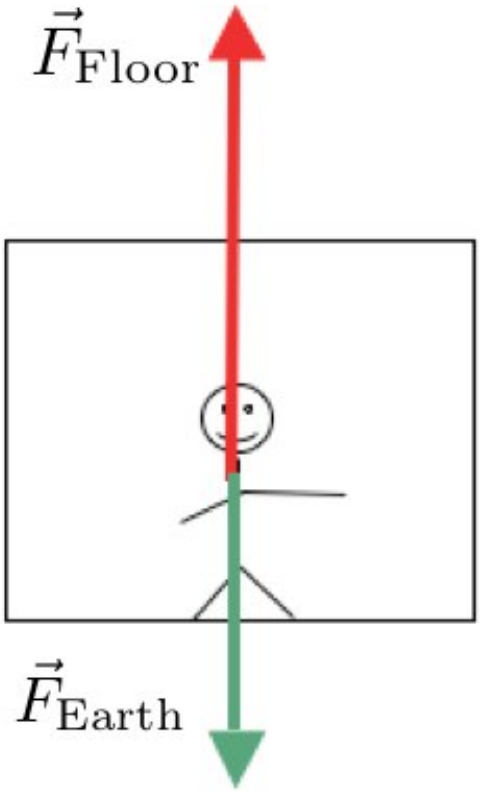

In terms of forces, what has to be different? If the person is accelerating upwards, the net force must also be upwards. Using the same two forces as above, there are two ways this can happen. The floor can push MORE on you, or the Earth can pull LESS. Since the gravitational force depends on your mass, the Earth's mass and the distance between those, it doesn't change. This means that the floor must push harder on you. But wait, you feel heavier and yet the gravitational force is the same.

Example 3: You are nearing the top floor and the elevator has to stop. Since it was moving up, but slowing down it has to accelerate in the downward direction.

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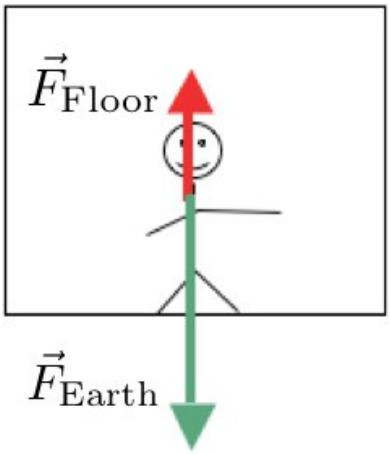

Now the net force must be in the downward direction. Again, the magnitude of the gravitational force doesn't change. The only thing that can happen is for the floor to push less. From this, you feel lighter. Right?

Last Example: Suppose the elevator cable breaks and the elevator falls. In this case, the acceleration of the elevator will be -9.8 m/s2 (just like any free falling object). How much would the floor have to push up on the person to accelerate down at -9.8 m/s2? It wouldn't have to push at all. The force the floor exerts on you would be zero. How would you feel? You would feel scared - I mean you are in an elevator with the cable cut. How else do you think you would feel? Well, maybe you could be scared AND hungry if you were late for lunch or something. Oh, you would feel weightless. Could this really happen? Absolutely. In fact, some people even pay to do this. Check out this ride, Superman:

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The basic idea is that you get in the car, it zooms up the vertical part of the track. During both the going up and going down parts fo the motion, the acceleration is -9.8 m/s2 so you feel weightless. Let me summarize so far:

In all these situations, the gravitational force does not change.

For the different situations, you have different accelerations.

The less the floor pushes on you, the lighter you feel.

If the floor doesn't push on you at all, you feel weightless.

Oh, there is another great example of this weightlessness on Earth. The vomit comet. Yes, it's real. Basically, it is a plane that flies in a manner that it has a downward acceleration the same as a free falling object. Just like the falling elevator except it doesn't hit the ground.

One more cool thing about the vomit comet. In the movie Apollo 13, the weightless scenes were filmed inside the vomit comet. This way, it wouldn't only look weightless, it would BE weightless. Of course, this means that they had to shoot scenes like 30 seconds at a time.

Back to the astronauts

The astronauts are in the Space Shuttle and the Space Shuttle is in orbit around the Earth. But is it accelerating? Yes. It is accelerating because the Earth pulls on it through the gravitational force. Even though it is moving in a circle, it is still accelerating. You could say the Space Shuttle is indeed falling since its motion is determined by the gravitational force. However, since it doesn't really get closer to the Earth during its motion it would be better to call it "in orbit". Think of this. Suppose you tie a string to a ball and swing it around your head in a near horizontal circle. Does the ball moving in a circle accelerate? Yes. If it accelerates, it must have a force in the direction of the acceleration. For the ball, this would be the tension in the string that pulls it towards the center of the circle. For an orbiting object, the gravitational force pulls on spacecraft. Well, what if you take a giant ball and string and swing it around. If you put a person inside the ball, would that person be weightless? No. The difference with gravity is that it pulls on all parts of spacecraft and all parts of a person's body. If you were in a giant circular moving ball, the wall of the ball would have to push on you. Maybe this diagram will help.

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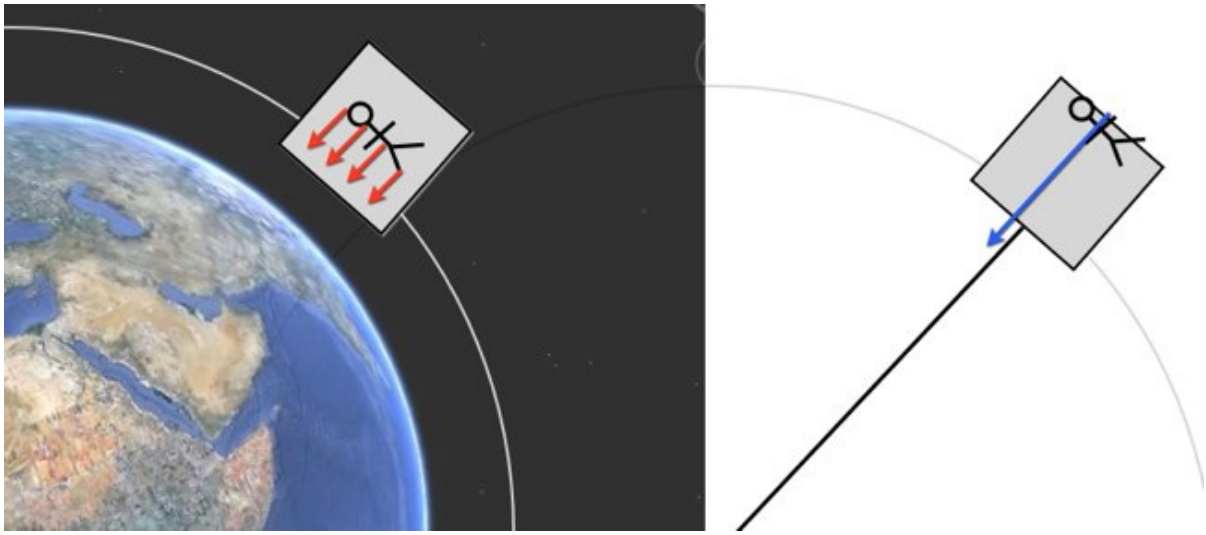
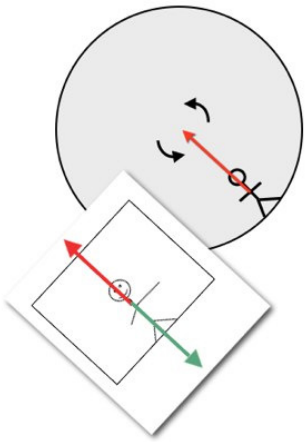


But what if you are actually in a place where the gravitational force is zero (like far away from other massive objects)? Can you make it feel like you have weight in this case? Yes. This is essentially the opposite of the orbital case. If you can make the spacecraft accelerate with a magnitude of 9.8 m/s2, it will feel just like you are on Earth. One way to accelerate would be with rockets. Maybe this would be a useful thing if you are trying to get to another star or something because you would get faster and faster. But what if you don't really want to go anywhere, but you want to feel like it does on Earth? Well, you could make a spaceship that spins. By moving in a circle (on the inside of the spacecraft), you would have an acceleration and thus a net force. Here is an astronaut in a spinning space ship in a region with no gravity. Next to that astronaut, I put a diagram of a person in an elevator. In both of these cases, the floor pushes up on the astronaut with the same magnitude. These two people would essentially feel the same (but not quite since the top fo the rotating head of the astronaut is actually moving differently than the feet).



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